MediLens AI | Diagnostic Imaging Support System

Center for Explainable Medical AI & Vision Transformers • B.Tech CSE (AI & ML) Capstone

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Status: Official Clinical Report

EXPLAINABLE PULMONARY DIAGNOSIS REPORT

Patient Name: Anitha Reddy	**Age / Gender:** 34 Yrs / Female	**Modality:** Digital Chest X-Ray (PA)
Patient ID: PID-00004	**Model:** Vision Transformer (ViT-B/16)	**XAI Method:** Attention Rollout

AI Diagnosis:	**PNEUMONIA DETECTED (Focal Consolidation)**	**Confidence: 98.5%**	**Pneumonia Prob:** 98.5%

Explainable AI Findings & Localization:

Affected Anatomical Zone: Left Mid-Zone Lobe (100.0% transformer attention focus)

Vision Transformer Multi-Head Attention identified prominent hyper-dense focal opacities and alveolar consolidation localized primarily in the Left Mid-Zone Lobe. The self-attention rollout mechanism reveals elevated activation weights across transformer encoder layers corresponding to classic radiological signs of pneumonia, including patchy airspace shadowing.

PHYSICIAN CLINICAL VERIFICATION & NOTES:

Attending Physician: Dr. Rajesh Sharma (MD Pulmonology, Senior Consultant) | **Reviewed:** 2026-09-23 05:34:27

Clinical Verdict: Confirmed Community-Acquired Pneumonia

Physician Notes: Radiological findings strongly corroborate the Vision Transformer self-attention localization. Right lower lobe airspace consolidation with subtle air bronchograms is evident. Heart size and mediastinal contours are preserved. No gross pleural effusion noted.

Recommendations: 1. Prescribed oral Azithromycin 500mg OD x 5 days with Paracetamol SOS for fever. 2. Incentive spirometry and adequate oral hydration. 3. Repeat chest radiograph in 14 days if symptoms do not resolve.

RESEARCH NOTICE & DISCLAIMER: *Educational & Research Prototype: MediLens AI is an academic research system designed for medical image classification using Vision Transformers and Explainable AI. Predictions and attention heatmaps generated by this tool are supportive indicators and do NOT constitute a definitive medical diagnosis. Always consult a qualified radiologist or healthcare provider for clinical medical interpretation.*